
CY 746: Self assembly of surfactants and Polymers in Solution

Course Code: CY 746

Course Name: Self assembly of surfactants and Polymers in Solution

L-T-P-C: 3-0-0-3

Prerequisites: Consent of the faculty member

Students intended for: Ph.D.

Elective or Core: Elective

Approval: 2nd Senate

Course Contents

Classification of surfactants, Basic and applied theories, surface active polymers, properties of surfactants, micelle formation, microemulsions, phase behaviour of surfactant systems, applications in nanomaterial synthesis, emulsion formation and stabilisation, foams, wetting, spreading and adhesion, personal care and cosmetics, pharmaceutical formulations, food industry

Text & Reference Books:

1. Milton J. Rosen, **Surfactants and Interfacial Phenomena**, Wiley Interscience.
2. K. Holmberg, B. Jonsson, B. Kronberg, B. Lindman, **Surfactants and Polymers in Aqueous Solution**, John Wiley & Sons Limited.
3. Richard J. Farn, **Chemistry and Technology of Surfactants**, Blackwell Publishing.
4. Tharwat F. Tadors, **Applied Surfactants: Principles and Applications**, Wiley-VCH.
5. John Texter, **Reactions and Synthesis in Surfactant Systems**, Marcel Dekker.